

Sanjit Kumar

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EDUCATION:

University of Illinois Urbana-Champaign, *Master of Computer Science*
2024

GPA: 3.91/4.00 | Aug 2022-May

Relevant Coursework: Distributed Systems, OS Design, Fault Tolerant Data Center Systems, ML and Data Systems, Artificial Intelligence, Data Mining, Advanced Software Engineering Topics, Interactive Computer Graphics.

Vellore Institute of Technology, *Bachelor of Technology in Computer Science and Engineering*
2022

GPA: 3.85/4.00 | Jul 2018-May

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Object Oriented Programming, Web Programming, Computer Networks, Graph Theory and Social Networks, Parallel Programming.

SKILLS:

- Programming Languages**: Python, Java, C++, C, Javascript, Go
- Databases**: MySQL, MongoDB, Redis, Elasticsearch
- Web Technologies**: ReactJS, Node.js, Express.js, PHP
- Other Technologies**: Apache Kafka, Hadoop, Kibana, Docker, AWS, Linux Kernel, Bash, Data Plane Development Kit (DPDK), OpenGL, WebGL, PyTorch, Grafana, JavaFX.

EXPERIENCE:

Amazon Music, *Software Development Engineer*

Java | Sep 2024 – Present | Seattle, WA

- Worked on multiple experimental features and A/B tests to drive Signups on the Amazon Music Growth – Tier Progressions team.
- Major experiments include contextualizing upsells for mobile users in Android, understanding pacing for Alexa users and leveraging proactive notifications to target specific offer eligible customers via Echo devices.

University of Illinois Urbana Champaign, *Graduate Teaching Assistant*

C++, Go | Jan 2024 – May 2024 | Champaign, IL

- Worked with Dr. Radhika Mittal on the course [CS425: Distributed Systems](#) to provide industry relevant lessons for 200+ graduate/undergraduate students. Held bi-weekly office hours and assessed homework and projects.
- Mentored students to build distributed & networked application projects (reliable multicasting, distributed transactions, raft).

Aviz Networks Inc, *Fall Software Engineer Intern*

C, Python, Redis | Sep 2023 – Nov 2023 | San Jose, CA

- Developed and integrated a UI, REST API, and Redis DB Cache for a C-based Network Packet Collector for GTP Correlation.
- Worked with kernel-bypass strategies via Data Plane Development Kit to achieve packet processing speed for datacenter networks.
- Integrated an Apache Kafka module using Kafka-C drivers into the packet data processing pipeline, enhancing reliability.
- Optimized and performance-tuned the packet collector pipeline via payload batching. Achieved a 3x improvement in throughput and 50% reduction in latency.
- Automated the configuration, build and execution of the packet collector with Bash scripts and Python.

Aviz Networks Inc, *Summer Software Engineer Intern*

C++, Python | May 2023 – Aug 2023 | San Jose, CA

- Designed and developed a C++ based scalable network packet analyzer system. System tapped packets from high throughput network traffic from data centers to extract and stream metadata information.
- Leveraged scalable event processing systems like Kafka and Elasticsearch for data pipelining and downstream analytics.
- Benchmarked scalability on physical network devices with software-based (Scapy) and hardware-based (Ixia) load generators.
- Used a Kafka consumer to integrate a REST API with the system for selective packet capture and data sink integrations.

WebKnot Technologies, *Full Stack Developer Intern*

ReactJS, Node.js | Nov 2020 – Dec 2020 | Bangalore, India

- Integrated custom Tensorflow-based object detection models with Shinobi, an open-source CCTV framework.
- Developed UI with ReactJS and wrote REST API endpoints with node.js and Express.js for two different MERN stack applications.

PROJECTS:

Distributed ML Inference Job Scheduler

[GitHub](#) | Java, Python, PyTorch

- Designed a distributed job scheduler system for ML inference tasks built on top of 10 Linux VMs from scratch using Java and Python.
- Uses a real-time work scheduling algorithm to optimize query rate for ResNet and ImageNet classification tasks.
- Includes a distributed data logging service, distributed group membership protocol and failure detector, a distributed files system.

Online Newsletter – The Weekly Edge

[GitHub](#) | Node.js, ReactJS, MongoDB

- A full scale MERN stack web application for a weekly online newsletter for the Journalism Club, VIT University. Complete workflow from writing the articles to reviewing, editing and publishing on public website.

WildSprint: Wildlife Fundraising Platform (*1st Place Ethereum Track – DevSpace '20 Hack*)

[GitHub](#) | Node.js, ReactJS, Solidity

- Developed a MERN stack fundraising platform to raise funds via cryptocurrency (wrote smart contracts to receive ETH coin).
- Built a live stream feature through [Dyte SDK](#) integration. Livestreams of wildlife in national parks to incentivize donations. [Demo](#).

RESEARCH EXPERIENCE:

Internet of Things Security: Attacks, Solutions, Strengths and Limitations

[Link](#) | Sep 2021

International Conference on Artificial Intelligence and Machine Vision, IEEE

- Presented a comparative analysis of benchmarks between latest security frameworks in then-recent IoT literature while advised by Dr. Anil Kumar Kakelli. Categorically classified and critiqued existing IoT security frameworks based on their approaches to address the threat of malignant nodes in heterogenous device networks and general strengths/limitations.